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SPECIALIZED PROJECT

**COSMO: an AI-native system for a connected
universe of customer information and related
actions in commerce**

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1 Introduction

1.1 System Description

COSMO is a campaign management system for structured outreach to external stakeholders (e.g., organizations and decision-makers) in B2B-style scenarios such as product introduction, partnership proposals, and customer acquisition. In these settings, outreach is not a single message but a multi-step process (initial contact, follow-up, scheduling, clarification) that requires both consistent execution and continuous context reuse.

1.1.0.1 Context and motivation. In practice, outreach often becomes inefficient and ineffective when it is executed manually across fragmented tools (email threads, spreadsheets, personal notes). This fragmentation creates three recurring issues: (1) *context loss* (teams cannot reliably reuse prior interaction history), (2) *execution inconsistency* (follow-ups depend on individual memory and vary by team member), and (3) *limited measurability/auditability* (it is hard to answer what actions were taken for each contact and which steps drive outcomes). These issues matter because outreach performance depends strongly on timing, consistency, and personalization; missing or delayed follow-ups can directly reduce reply and conversion rates, while manual work increases operational cost per lead.

1.1.0.2 Rationale from related systems (commercial and open-source). Commercial platforms such as HubSpot and Salesforce Marketing Cloud demonstrate a widely adopted paradigm: outreach automation works best when it is driven by CRM data and executed through explicit workflows/journeys rather than ad-hoc messaging [3, 11]. Sales engagement systems operationalize a similar idea via sequences and prioritized tasks (repeatable steps executed at scale) [10]. Open-source marketing automation suites (e.g., Mautic) further show that workflow-based campaigns and segmentation are useful even in self-hosted environments [7]. These systems collectively motivate COSMO’s central abstraction: *a campaign is a workflow over a well-defined audience, executed step-by-step, and recorded as auditable events*.

1.1.0.3 Why a workflow-execution model (task queue + agents). To address reliability and traceability, COSMO models each campaign step as a queued task processed by execution agents, with explicit states (`queued` → `processing` → `succeeded/failed`). This enables retries, monitoring, and auditable history, aligning with established workflow management principles where fault tolerance and accountability are achieved via explicit execution semantics [1]. Agents decouple campaign logic from core data management, making the system extensible to new channels or behaviors without rewriting the entire backend.

1.1.0.4 Why AI-assisted outreach (summarization + next-best-action + drafting). Beyond deterministic automation, COSMO includes intelligence agents to reduce cognitive load and increase personalization quality. They (1) summarize prior exchanges for quick context, (2) recommend next-best actions (e.g., follow up, schedule a call, pause) following the “next best action” paradigm used in modern CRM systems [12], and (3) generate grounded message drafts using CRM fields and interaction summaries, similar to AI-assisted email drafting features in industry tools [4, 8]. However, academic findings indicate that automated lead nurturing can



improve engagement but is not universally beneficial; therefore, COSMO supports feedback mechanisms and optional human validation of AI-generated insights [5, 2].

1.1.0.5 Architecture and authentication. COSMO follows a client-server architecture: the web frontend provides management UI and visualization, while the backend exposes REST APIs for business logic, persistence, and orchestration; background workers run execution and intelligence agents. Authentication uses Google OAuth with the authorization flow initiated from the frontend client; the backend handles token exchange, issues JWT access/refresh tokens, and enforces organization-scoped authorization.

1.2 Objectives

The primary objectives of COSMO are to make outreach scalable, consistent, and traceable by unifying contact context with reliable campaign execution. Specifically, COSMO aims to:

1. **Organization-scoped access and governance:** support authentication and role-based authorization, and enforce strict data separation between organizations (multi-tenant access control).
2. **Single source of truth for contacts:** maintain a unified contact profile that includes organization context, customizable fields, and a timeline of collected information and interactions, so any team member can immediately understand what is known and what has already been communicated.
3. **Audience segmentation and list management:** allow users to build and maintain contact lists/segments that can be reused across campaigns and updated as contact data changes.
4. **Campaign workflow management:** define campaigns over a target list with configurable cadence, message strategy (templates/policies), and status tracking (e.g., draft/active/paused/completed) to ensure consistent execution.
5. **Reliable task orchestration and monitoring:** execute campaign steps as background tasks (e.g., send email, wait, follow up, schedule meeting) with explicit task states, retries, and execution logs to support monitoring, auditability, and coordination across team members.
6. **AI-assisted outreach (optional):** provide automated interaction summarization, next-best-action suggestions, and grounded message drafting to reduce repetitive work while preserving personalization, with policy gating and optional human review for sensitive or high-risk outreach.
7. **Extensible system architecture:** implement a clean client-server separation (frontend UI, REST backend, background workers/agents) and modular domain boundaries so new channels, agents, and features can be added without major refactoring.

1.3 Stakeholders

The main stakeholders involved in this project include:



- **Project developers:** responsible for system design, implementation, and testing.
- **Supervising instructor:** evaluates the technical design, implementation quality, and project outcomes.
- **Intended users:** sales or marketing personnel who use the system to manage contacts and outreach campaigns.
- **Organization administrators :** manage organizational data and user configurations.

1.4 Users

COSMO serves a single primary user type: **Business Development (BD) representatives**. BD users perform two core workflows: *automated email campaigns* and *AI-assisted manual outreach*. Users are grouped into **organizations**, and each user holds one of two roles within an organization:

- **Admin:** Full access to all organization data. Admins can update organization settings, invite and manage members, and assign roles. The creator of an organization is automatically assigned the Admin role.
- **Member:** Can manage contacts, contact lists, campaigns, templates, agents, and tasks within the organization, but cannot modify organization settings or manage other members.

Both roles share the same operational capabilities for day-to-day BD work (contact management, campaign execution, outreach). The role distinction applies only to organization administration.

Users can create and manage **execution agents**, which represent email accounts connected via OAuth (e.g., Gmail) and used as sending identities for outreach. Agents are not interactive users; instead, background workers use these agent credentials to execute campaign tasks on behalf of the owning user.

1.5 Problem Statement

Many organizations need to conduct structured outreach to external decision-makers in order to introduce a product, propose a collaboration, or acquire customers. For example, a company may want to contact event organizers, venue operators, or association representatives to present its solution and gradually move them toward a response or meeting. In practice, this process is often executed manually through scattered email threads, spreadsheets, and personal notes, which turns outreach into a real operational problem rather than a minor inconvenience. In particular, there is no shared “single source of truth” for each contact: the team cannot quickly see what information has been collected and what has already been communicated, leading to duplicated effort and context loss.

When outreach relies on manual effort, team members must repeatedly search for contact context, reconstruct what has been discussed, decide the next step, and write messages from scratch. As the number of contacts grows, follow-ups are easily missed or delayed, messaging becomes inconsistent across different team members, and it becomes difficult to maintain personalization and timing—all of which directly reduces reply rates and conversion. Moreover, without a unified execution workflow and reliable activity logs, teams cannot clearly answer what actions were taken for each contact, measure which campaign steps are effective, or iterate

systematically to improve outcomes. Therefore, the core problem is the difficulty of scaling B2B-style product introduction outreach while maintaining speed, consistency, accountability, and effectiveness.

1.6 Challenges

Despite being a common business need, structured outreach remains difficult to execute effectively at scale due to several inherent challenges:

- **Fragmented context and ownership:** contact information and interaction history are typically spread across email threads, spreadsheets, and individual notes, so teams lack a shared single source of truth and quickly lose context when members change or workloads grow.
- **Consistency vs. personalization trade-off:** outreach needs to follow a consistent process (cadence, follow-ups, messaging standards) while still being personalized for each organization/contact; manual personalization does not scale, but overly generic automation reduces reply rates.
- **Timing and follow-up discipline:** effective outreach depends heavily on timely follow-ups and clear next steps; when outreach is manual, tasks are easily delayed or forgotten, especially when managing many contacts in parallel.
- **Traceability and accountability:** teams need to know what has been sent, what happened, and what should happen next for each contact. Without explicit tracking, it is hard to audit actions, avoid duplicate outreach, and coordinate across team members.
- **Measuring and improving effectiveness:** outreach quality is hard to improve without reliable records of steps and outcomes (e.g., which messages/cadences work). Many teams cannot iterate systematically because execution data is incomplete or inconsistent.
- **Risk and governance in automation:** automating messages can introduce reputational, compliance, or relationship risks. Teams often require policy constraints and human review for sensitive segments or high-stakes communications, which makes fully automatic systems harder to deploy safely.

1.7 Reasons for Implementation

COSMO is implemented to address a practical gap in B2B-style outreach: teams often need to introduce a product or proposal to external organizations and decision-makers, but the process becomes inefficient and unreliable when contact context and interaction history are scattered across multiple tools, and when follow-ups depend on manual discipline. This fragmentation reduces visibility (it is unclear what has been collected and what has already been communicated), causes missed or delayed follow-ups, and prevents teams from measuring and improving outreach effectiveness systematically.

Motivated by the workflow-based approach commonly seen in modern outreach platforms (CRM-driven segmentation, campaigns, and step-by-step execution), COSMO is designed as a unified system that (1) provides a single source of truth for each contact (profile, organization context, and interaction timeline), and (2) executes campaigns as traceable workflows through a task queue and agents. This execution model makes campaign progress observable and auditable



by recording explicit task states (e.g., queued, processing, succeeded/failed) and execution logs, enabling reliable follow-ups, coordination across team members, and iteration based on outcomes.

In addition, COSMO is implemented to explore AI-assisted outreach as a practical extension of automation: summarizing prior exchanges, recommending next best actions, and drafting grounded personalized messages to reduce repetitive work while maintaining personalization quality. Finally, COSMO serves as a real-world-inspired case study for applying software engineering principles—modular design, layered client–server architecture, clear domain separation, and background worker orchestration—in a system that must remain extensible as features evolve.

2 Literature Review

Table 1: Direct comparison of COSMO with related commercial and open-source outreach systems

Capability	COSMO	HubSpot	SFMC	Outreach	Mautic	Odoo MA	SuiteCRM	listmonk
Contact database / CRM	✓	✓	✓	✓	✓	✓	✓	partial
Organization / account context (B2B)	✓	✓	✓	✓	partial	partial	✓	✗
Custom fields / contact enrichment	✓	✓	✓	✓	✓	✓	✓	partial
Segments / contact lists	✓	✓	✓	✓	✓	✓	✓	✓
Campaign workflows / journeys	✓	✓	✓	partial ^a	✓	✓	partial	✗
Task/sequence execution model (cadence, follow-up)	✓	partial	partial	✓	partial	partial	✗	✗
Execution state tracking + audit logs	✓	partial ^b	partial ^b	partial	partial	partial	partial	partial
AI assistance (summary / NBA / drafting)	✓	partial	partial	partial	✗	✗	✗	✗
Self-hosted / on-premise	✓	✗	✗	✗	✓	✓	✓	✓
Extensibility (APIs / integrations / plugins)	✓	✓	✓	✓	✓	✓	✓	✓

^a Outreach focuses on sales engagement sequences rather than full marketing journeys.

^b Commercial platforms provide automation logs, but low-level queue/worker details are typically abstracted from end users.

Sources: HubSpot [3], Salesforce Marketing Cloud (Journeys) [11], Outreach [10], Mautic [7], Odoo Marketing Automation [9], SuiteCRM Campaigns [13], listmonk [6].

3 Business Requirements

Business requirements define the high-level organizational needs that COSMO must satisfy, independent of technical implementation.

BR-01 The system shall enable Business Development (BD) representatives to manage the full outreach lifecycle—from identifying target contacts, through initial outreach and follow-ups, to meeting scheduling and outcome tracking—within a single platform.

BR-02 The system shall support two distinct outreach workflows:

- **Automated email campaigns:** The system sends emails in bulk via connected email accounts and automatically schedules follow-ups according to a configured cadence.
- **AI-assisted manual outreach:** The system suggests prioritized contacts, generates personalized draft messages, and tracks interaction outcomes, while the BD representative sends messages manually through external channels (LinkedIn, personal email, phone).



- BR-03** The system shall support multi-tenant data isolation, where each organization operates as an independent workspace with its own contacts, campaigns, agents, and configuration. Users must not be able to access data belonging to other organizations.
- BR-04** The system shall enforce role-based access within each organization:
- **Admin:** Full access including organization settings, member management, and role assignment.
 - **Member:** Full access to all BD operational functions (contacts, campaigns, agents, outreach) but cannot modify organization settings or manage members.
- BR-05** The system shall leverage AI (via OpenAI GPT-4) to reduce manual effort in content creation, including: generating personalized email templates for campaigns, generating outreach draft messages tailored to contact context and scenario, enriching contact profiles with structured insights, and producing pre-meeting briefings.
- BR-06** The system shall offload long-running operations (email delivery, campaign execution, credential synchronization, score recalculation) to background workers so that user-facing interactions remain responsive.
- BR-07** The system shall provide a human-in-the-loop feedback mechanism where BD representatives can rate AI-generated suggestions (used as-is, modified, wrote own, or skipped) and record outreach outcomes (replied, meeting booked, dropped), enabling measurement of AI effectiveness per scenario.
- BR-08** The system shall capture inbound leads via public-facing forms that external visitors can submit without authentication, creating contact records that feed into the campaign pipeline.
- BR-09** The system shall support programmatic access via personal API keys and external AI tool integration (Model Context Protocol), enabling third-party automation tools to create campaigns with contacts through a single API call.

4 Data Requirements

Data requirements specify the data entities, structures, constraints, and policies that govern how COSMO stores and manages information.

4.1 Core Data Entities

The system manages the following primary entities, organized by domain:

- **User & Identity:** Users, Roles, Personal API Keys, Integrations (OAuth credentials).
- **Organization:** Organizations with configurable settings.
- **Contact Data:** Contacts, Custom Fields, Contact Lists (grouping), Inbound Lead Forms.
- **Campaign & Content:** Campaigns, Templates (ordered sequences), Emails (sent records), Conversations (thread history).



- **Execution:** Agents (email accounts), Tasks (execution units).
- **AI-Assisted Outreach:** Outreach States, Interaction Logs, Meetings, Segmentations, Contact-Segment Scores, Playbooks, Automation Rules, Enrollments, User Feedback.

4.2 Flexible Schema via JSONB

To support evolving business requirements without frequent database migrations, the system stores semi-structured data in PostgreSQL JSONB columns:

- **Contact profile & insights:** `profile` (social links, contact info), `ai_insights` (AI-generated analysis), `confirmed_facts` (human-validated data), `scores` (relationship and segment scores), `tags` (categorization). The `profile` column uses a GIN index for efficient querying.
- **Campaign configuration:** `outreach_strategy` (positioning, tone), `follow_up_schedule` (cadence timing), `client_metadata` (client-facing settings), `playbook` (sequence definition).
- **Task payload & result:** `payload` (agent ID, template info, contact data), `result` (execution outcome, error details).
- **Segmentation criteria:** `criteria` (filter rules), `icp_definition` (Ideal Customer Profile), `score_breakdown` (scoring rationale).
- **Credentials:** `credentials` (encrypted OAuth tokens for agents and integrations).

4.3 Data Constraints

4.3.0.1 Uniqueness constraints. The system enforces the following composite unique constraints to prevent duplicate records:

Entity	Unique Constraint
User	<code>email</code> (global)
Role	<code>(user_id, organization_id)</code>
Agent	<code>(user_id, email)</code>
Contact	<code>(source_id, source)</code>
Template	<code>(user_id, campaign_id, type)</code>
Task	<code>(contact_id, campaign_id, template_id)</code>
Custom Field	<code>(organization_id, name, entity_type)</code>
Outreach State	<code>(user_id, contact_id)</code>
Contact-Segment	<code>(contact_id, segmentation_id)</code>
Lead Form	<code>slug</code> (global)



4.3.0.2 Soft delete. All major entities implement soft deletion using an `is_deleted` boolean flag (default `false`, indexed) and a `deleted_at` timestamp. Records are never physically removed; queries filter by `is_deleted = false` by default. This preserves referential integrity and supports audit trails.

4.3.0.3 Enumerated values. The system uses string-typed enums validated at the application layer:

- Contact: `outreach_stage` (COLD, NO_REPLY, REPLIED, POST_MEETING, DROPPED), `lifecycle_stage`, `business_stage`.
- Campaign: `status` (draft, active, paused, ended, scheduled).
- Task: `status` (pending, running, done, failed, cancelled), `priority` (low, medium, high, critical).
- Agent: `status` (active, inactive, insufficient_scopes, needs_sync_setup, invalid_google_grant).
- Meeting: `status` (scheduled, completed, cancelled, no_show).
- Role: `name` (admin, member), `status` (pending, active).
- Outreach: `next_step` (SEND, FOLLOW_UP_1, FOLLOW_UP_2, SET_MEETING, PREPARE_MEETING, WAIT, DROP).
- Custom Field: `data_type` (text, number, email, select, date, url).

4.4 Data Isolation

All organization-scoped queries enforce tenant isolation by including an `organization_id` filter. The repository layer uses the pattern:

```
WHERE (user_id = ? OR organization_id = ?) AND is_deleted = false
```

This dual-scoping allows entities to exist at either user level (`organization_id` is NULL) or organization level, while preventing cross-tenant data access.

4.5 Custom Field Extensibility

The contact schema supports runtime extensibility via the Custom Fields module. Custom field definitions specify a `name` (normalized to snake_case), `data_type`, and optional `options` (for select-type fields). Custom field values are stored as JSONB within the contact record, eliminating the need for database schema migrations when adding new fields.

5 System Requirements

5.1 Functional Requirements

The functional requirements are organized by module, reflecting the system's domain-driven architecture. COSMO uses a single user entity with organization-scoped roles (**admin** and **member**). Resource access is governed by three patterns: (1) authentication only, (2) resource ownership verification, and (3) organization role checks.

Requirements are divided into two phases:

- **Phase 1 (Core Platform):** Fully operational modules with registered API routes and frontend UI.
- **Phase 2 (AI-Assisted Outreach):** Modules with complete backend implementation (handlers, services, domain models, database migrations, background workers) pending API route activation.

5.1.1 Phase 1: Core Platform

5.1.1.1 Authentication and User Account Management. These operations are available to any user, with or without organization membership.

- **FR-01:** The system shall allow a user to authenticate via Google OAuth 2.0. The backend exchanges the authorization code for tokens, creates or retrieves the user record, and issues a JWT access token and refresh token.
- **FR-02:** The system shall allow a user to obtain a new access token by presenting a valid refresh token, without requiring re-authentication.
- **FR-03:** The system shall allow a user to log out, invalidating the current session.
- **FR-04:** The system shall allow an authenticated user to retrieve, update, and delete their own profile.
- **FR-05:** The system shall allow an authenticated user to create, list, and revoke personal API keys as an alternative authentication method for programmatic access.

5.1.1.2 Organization Management. The user who creates an organization is automatically assigned the **admin** role. Only administrators may modify organization settings or manage members.

- **FR-06:** The system shall allow an authenticated user to create a new organization. The creating user is automatically assigned the **admin** role.
- **FR-07:** The system shall allow an authenticated user to list all organizations in which they hold an active role.
- **FR-08:** The system shall restrict organization updates (name, settings) to users with the **admin** role in that organization.
- **FR-09:** The system shall restrict adding members and assigning roles to users with the **admin** role. Administrators may assign either **admin** or **member** roles to new members.

5.1.1.3 Contact Management. Contact data is scoped by organization. Operations are available to any authenticated organization member.

- **FR-10:** The system shall allow an authenticated user to create, view, update, and delete contacts within their organization.
- **FR-11:** The system shall support searching contacts by keywords and filtering by structured attributes including custom fields.
- **FR-12:** The system shall expose an endpoint to retrieve distinct filterable field values, enabling dynamic filter construction on the frontend.
- **FR-13:** The system shall support bulk deletion of contacts.
- **FR-14:** The system shall allow an authenticated user to create, view, update, and delete custom field definitions that extend the contact schema without requiring database migrations.

5.1.1.4 Contact List Management. Contact lists group contacts into reusable audiences for campaigns.

- **FR-15:** The system shall allow an authenticated user to create, view, update, and delete contact lists.
- **FR-16:** The system shall support searching contact lists by keywords and metadata.

5.1.1.5 Campaign Management. Campaigns define outreach workflows over a target contact list. Campaigns are scoped by organization.

- **FR-17:** The system shall allow an authenticated user to create, view, update, and delete campaigns.
- **FR-18:** The system shall support searching campaigns by keywords and status.
- **FR-19:** The system shall allow a user to assign an execution agent and team members to a campaign.
- **FR-20:** The system shall allow a user to update campaign status (e.g., draft, active, paused, completed).
- **FR-21:** The system shall allow a user to define an outreach strategy for a campaign, including message positioning and tone.
- **FR-22:** The system shall allow a user to configure a follow-up schedule for a campaign, defining the cadence and timing of follow-up steps.
- **FR-23:** The system shall allow a user to trigger AI-assisted generation of personalized email templates for a campaign, using contact context and the defined outreach strategy.

5.1.1.6 Agent Management. Agents are execution entities that send emails via connected email accounts. Access follows an ownership model: only the owner may update, delete, or synchronize an agent; organization members may view agents within the same organization.

- **FR-24:** The system shall allow an authenticated user to create an execution agent linked to their email account via OAuth (e.g., Gmail).
- **FR-25:** The system shall allow the agent owner to view, update, and delete their agents. Organization members may view agents belonging to the same organization.
- **FR-26:** The system shall support searching agents by keywords and status.
- **FR-27:** The system shall allow a user to search and view conversation history between a specific agent and its contacts, subject to ownership or organization membership.

5.1.1.7 Template Management. Templates define reusable email content for campaign sequences. Only the template creator may update or delete a template.

- **FR-28:** The system shall allow an authenticated user to create, view, update, and delete message templates. Update and delete operations are restricted to the template owner.
- **FR-29:** The system shall support associating templates with campaigns and ordering them in a sequence for multi-step outreach.

5.1.1.8 Email Record Viewing. Sent emails are recorded for auditability. Access is scoped to organizations where the user holds an active role.

- **FR-30:** The system shall allow an authenticated user to view sent email records, filtered to organizations in which the user holds an active role.

5.1.1.9 Task Orchestration. Tasks represent individual execution steps within campaign workflows. The system provides both direct task management and an asynchronous enqueue mechanism for background processing.

- **FR-31:** The system shall allow an authenticated user to create tasks and track their status through explicit state transitions: `queued` → `processing` → `succeeded` or `failed`.
- **FR-32:** The system shall provide API endpoints to enqueue tasks for asynchronous processing by background workers, including:
 - sending outreach emails via the assigned agent's email account,
 - executing campaign workflows (multi-step email sequences over a contact list),
 - synchronizing agent credentials and OAuth state,
 - scheduling future follow-up tasks according to the campaign cadence.
- **FR-33:** The system shall allow an authenticated user to view task details and list tasks, optionally filtered by campaign or contact.



5.1.1.10 Inbound Lead Forms. Inbound lead forms allow external visitors to submit contact information. Form management requires authentication; form submission is intentionally public to support external lead capture.

- **FR-34:** The system shall allow an authenticated user to create, view, update, and delete inbound lead forms. Update and delete are restricted to the form owner.
- **FR-35:** The system shall allow unauthenticated external users to submit an inbound lead form by its public identifier, creating a contact record and optionally triggering campaign-related workflows.

5.1.2 Phase 2: AI-Assisted Outreach

Phase 2 extends the core platform with intelligence, outreach orchestration, and automation capabilities. All modules described below have complete backend implementations—including HTTP handlers, services, domain models, database migrations, and background workers—and are pending API route registration.

5.1.2.1 Contact Intelligence. The intelligence module provides AI-powered contact analysis and scoring. It uses OpenAI for insight generation and Redis vector store for semantic search.

- **FR-36:** The system shall allow a user to trigger AI enrichment for a contact, generating structured insights (e.g., pain points, buying signals, communication preferences) and vector embeddings for semantic search.
- **FR-37:** The system shall allow a user to calculate segmentation fit scores for a contact against one or more defined segments.
- **FR-38:** The system shall allow a user to compute a relationship strength score for a contact based on interaction history, using weighted metrics (reply frequency, meeting count, engagement recency over 30-day and 90-day windows).
- **FR-39:** The system shall allow a user to generate a pre-meeting briefing for a contact, including talking points, discovery questions, and risk flags, generated by AI using the contact's profile, interaction history, and segment fit data.
- **FR-40:** The system shall provide campaign-level intelligence by aggregating reply rates, response counts, and generating AI-powered recommendations for campaign optimization.

5.1.2.2 Vector Search. The vector search module enables semantic discovery of contacts, interactions, and segments using AI-generated embeddings stored in a Redis vector index (HNSW algorithm, cosine distance, 1536 dimensions).

- **FR-41:** The system shall support semantic vector search across contacts and interaction history using AI embeddings, with configurable result limits and similarity thresholds.
- **FR-42:** The system shall support finding contacts similar to a given contact based on embedding proximity.
- **FR-43:** The system shall support hybrid search combining keyword matching and semantic similarity for contact discovery.



5.1.2.3 Outreach State Management. The outreach module tracks the full lifecycle of contact engagement—from initial outreach through follow-ups, meetings, and outcomes.

- **FR-44:** The system shall suggest prioritized contacts for outreach (cold, follow-up, or mixed) based on pipeline state and interaction history. Administrators see all organization contacts; members see their own.
- **FR-45:** The system shall generate AI-powered outreach draft messages personalized to each contact's context, with configurable language (Vietnamese or English) and scenario (role-based, industry-based, no-reply follow-up, post-reply, meeting confirmation, post-meeting, re-engage). The system falls back to template-based drafts when AI generation is unavailable.
- **FR-46:** The system shall maintain an outreach state for each user-contact pair, tracking: conversation state (`COLD`, `NO_REPLY`, `REPLIED`, `POST_MEETING`, `DROPPED`), follow-up count, maximum follow-ups, days since last interaction, and recommended next step (`SEND`, `FOLLOW_UP_1`, `FOLLOW_UP_2`, `SET_MEETING`, `PREPARE_MEETING`, `WAIT`, `DROP`).
- **FR-47:** The system shall allow a user to record interactions (messages sent and received) with contacts, categorized by channel (Email, LinkedIn, Call, Meeting, Note), direction (outgoing, incoming, internal), and sentiment (positive, neutral, negative).
- **FR-48:** The system shall allow a user to add, update, and delete internal team notes for a contact.
- **FR-49:** A background worker shall periodically recalculate the outreach state and next step for contacts in `WAIT` state, updating the contact's outreach stage and follow-up count when state transitions occur.

5.1.2.4 Meeting Management. The meeting module supports scheduling, preparation, and outcome tracking for meetings with contacts.

- **FR-50:** The system shall allow a user to create, update, and delete meetings with contacts, including metadata such as title, time, duration, channel (Zoom, Google Meet, Call, Offline), location, and meeting URL.
- **FR-51:** The system shall track meeting status through a lifecycle: `scheduled` → `completed` or `cancelled` or `no_show`.
- **FR-52:** The system shall support AI-generated meeting preparation before a meeting and storage of meeting content (transcript or notes), outcomes, and next steps after a meeting.

5.1.2.5 Outreach Feedback and Analytics. The feedback system tracks how users interact with AI suggestions and measures outreach effectiveness per scenario.

- **FR-53:** The system shall record user actions on AI-generated suggestions: `used_draft`, `modified_draft`, `wrote_own`, or `skipped`.
- **FR-54:** The system shall record outreach outcomes per contact: `sent`, `replied`, `meeting_booked`, `meeting_done`, or `dropped`, with optional reply sentiment and days to reply.

- **FR-55:** The system shall provide aggregated feedback statistics including reply rate, meeting rate, draft usage rate, and average days to reply, grouped by outreach scenario and context level, via a precomputed database view.

5.1.2.6 Segmentation. The segmentation module enables dynamic contact grouping based on criteria and ICP (Ideal Customer Profile) definitions with automated scoring.

- **FR-56:** The system shall allow a user to create, view, and delete contact segments with configurable criteria (filters, scoring rules, exclusions), ICP definition, and priority (1–10).
- **FR-57:** The system shall maintain a fit score (0–100) for each contact–segment pair, computed automatically or set manually, with a detailed score breakdown. Scores track qualification status and campaign enrollment state.
- **FR-58:** The system shall support listing contacts within a segment with pagination and querying all segment scores for a given contact.
- **FR-59:** A background worker shall periodically recalculate segment scores for contacts in batches, using the intelligence service to evaluate contacts against active segment criteria.

5.1.2.7 Playbook Automation. Playbooks define multi-stage outreach sequences that can be executed automatically. The automation system connects segments to playbooks via trigger rules with optional human approval.

- **FR-60:** The system shall allow a user to create, view, and delete playbooks with configurable multi-stage sequences. Supported stage types are: email, LinkedIn, call, and wait. Each stage defines trigger conditions (wait duration in days), success criteria (advance, pause, or complete on timeout), and content (template text or AI generation prompt).
- **FR-61:** The system shall allow a user to create, view, toggle, and delete automation rules that link a segment to a playbook with enrollment criteria (e.g., minimum fit score and engagement score thresholds) and an optional human approval requirement.
- **FR-62:** The system shall support manual and automatic enrollment of contacts into playbooks. When an automation rule requires human approval, the system shall create an approval request with the contact's fit and engagement scores that must be approved or rejected by a reviewer before enrollment proceeds.
- **FR-63:** A background worker shall evaluate active automation rules by finding contacts that meet enrollment criteria in the rule's linked segment and are not yet enrolled, creating enrollments or approval requests as configured.
- **FR-64:** A background worker shall process active enrollments by executing the current stage when its wait duration has elapsed. For email stages, the worker identifies the contact's email, selects the user's default active agent, generates content (from template with placeholder substitution or AI generation), and enqueues an email-send task. After execution, the worker advances to the next stage or marks the enrollment as completed.
- **FR-65:** The system shall track enrollment status (`pending_approval`, `active`, `paused`, `completed`) and maintain an execution log recording which stages have been completed and when.

5.1.2.8 Interaction Logging. The interaction module provides a unified log of all contact interactions across channels, feeding into the intelligence and automation pipelines.

- **FR-66:** The system shall allow a user to log interactions with contacts, categorized by type (sent, open, reply, bounce, LinkedIn message, call, meeting), channel, direction, and optional AI analysis metadata.
- **FR-67:** The system shall list recent interactions for a contact with pagination.
- **FR-68:** When a new interaction is logged, the system shall enqueue an orchestration task that re-enriches the contact, recalculates segment scores, updates the relationship score, and evaluates playbook automation rules.

5.1.2.9 User Feedback on AI Insights. The feedback module enables users to validate, correct, or extend AI-generated contact insights, supporting a human-in-the-loop approach to data quality.

- **FR-69:** The system shall allow a user to record score adjustments (original vs. adjusted score with reason), insight validations (confirm or reject AI-generated insights with optional confirmed data), and custom facts for a contact.
- **FR-70:** The system shall list a user's feedback history for auditability.

5.1.2.10 MCP Integration (Model Context Protocol). The MCP module provides a simplified API for external AI tool agents to create campaigns with contacts in a single call.

- **FR-71:** The system shall allow an authenticated AI tool agent to create a campaign with an associated contact list and contacts in a single API call. The system creates or updates contacts by email, generates a campaign name from the playbook type if not provided, notifies the user, and triggers asynchronous AI template generation.

5.2 Non-Functional Requirements

The most critical architectural characteristics for COSMO are, in order of priority: **Responsiveness**, **Reliability**, **Maintainability**, and **Security**.

- **Responsiveness:** Users interact frequently with contact lists, search, and campaign monitoring. Slow UI or API responses reduce productivity and make campaign operations harder to manage.
- **Reliability:** Campaign execution depends on consistent APIs and stable background processing. Failures or inconsistent responses can cause missed outreach steps and reduce trust in the system.
- **Maintainability:** COSMO is expected to evolve with new features (e.g., integrations, AI-assisted content generation). A modular domain separation approach reduces coupling and supports long-term development scalability.
- **Security:** COSMO manages sensitive contact and campaign data in a multi-tenant environment. Strong authentication, token handling, and data isolation are required to prevent unauthorized access.

5.2.1 Responsiveness Requirements

- **NFR-01:** The system must provide a responsive web user interface with efficient client-side rendering using Next.js and server-side data caching via TanStack React Query.
- **NFR-02:** The system must support fast list rendering for large datasets via server-side pagination and optimized database queries with appropriate indexes.
- **NFR-03:** The system must return search and filter results within an acceptable time threshold under normal load (target: $< 500\text{ms}$ for paginated queries).
- **NFR-04:** The system must offload long-running operations (email sending, AI content generation, contact enrichment, score recalculation) to background workers so that API responses remain fast.

5.2.2 Reliability Requirements

- **NFR-05:** The system must provide consistent REST API responses with standardized error formats across all endpoints.
- **NFR-06:** The system must enforce authentication checks consistently via middleware on all protected endpoints, with explicitly defined public path exceptions (login, token refresh, lead form submission).
- **NFR-07:** The system must ensure queued tasks are processed reliably by background workers (Asyncq), with automatic retry on transient failures (network timeouts, rate limits) and explicit failure states when retries are exhausted.
- **NFR-08:** The system must preserve explicit task state transitions (`queued` $\rightarrow$ `processing` $\rightarrow$ `succeeded/failed`) to support monitoring, debugging, and auditability.
- **NFR-09:** The system must include health check endpoints (`/health`, `/healthz`, `/ping`) to support deployment monitoring and container readiness probes.

5.2.3 Maintainability Requirements

- **NFR-10:** The backend must follow a layered architecture with clear separation: HTTP handlers (controllers), services (business logic), and repositories (data access), organized by domain modules.
- **NFR-11:** The system must maintain a clear separation between the frontend (presentation and client-side logic) and backend (business logic and persistence), communicating exclusively through REST APIs.
- **NFR-12:** The system must support adding new domain modules and API endpoints with minimal impact on existing components, facilitated by a dependency injection pattern for handler, service, and repository initialization.
- **NFR-13:** The system must support API versioning (`v1`, `v2`, `v3`) to enable backward-compatible evolution of endpoints.
- **NFR-14:** The contact schema must support runtime extensibility via custom fields (stored as structured metadata) and JSONB columns (e.g., `ai_insights`, `confirmed_facts`, `scores`) without requiring database schema migrations.



5.2.4 Security Requirements

- **NFR-15:** The system must protect all non-public API endpoints using JWT-based authentication, validated through a global middleware layer.
- **NFR-16:** The system must support refresh tokens to maintain user sessions without frequent re-authentication, while enforcing token expiry.
- **NFR-17:** The system must enforce access control at the handler level using three patterns:
 - (a) authentication-only for general operations,
 - (b) resource ownership verification for user-scoped resources (agents, templates, lead forms),
 - (c) organization role checks (**admin** required) for organization management operations.
- **NFR-18:** The system must enforce organization-scoped data isolation to prevent cross-tenant data access. All organization-scoped queries must filter by the authenticated user's organization context.
- **NFR-19:** The system must support per-IP rate limiting on API endpoints to mitigate abuse.
- **NFR-20:** The system must support personal API keys with encrypted storage as an alternative to JWT tokens for programmatic access.

6 System Overview

6.1 System Analysis

6.1.1 Usecase Diagram

6.1.2 Sequence Diagram

COSMO involves five actors:

- **Admin:** A BD representative with the **admin** role within an organization. Has full access to organization data and administration features, including managing organization settings, adding/removing members, and assigning roles. The creator of a new organization is automatically assigned the Admin role.
- **BD Representative (Member):** A BD representative with the **member** role. Can perform all core business operations (managing contacts, campaigns, agents, and outreach) except for organization settings and member management.
- **External Visitor:** A non-authenticated user outside the system. Can only submit inbound lead forms through a public URL.
- **Background Worker (Asynq):** An automated system actor that processes the task queue (e.g., sending emails, executing campaigns), recalculates outreach states, recomputes segment scores, and executes playbook enrollment stages.

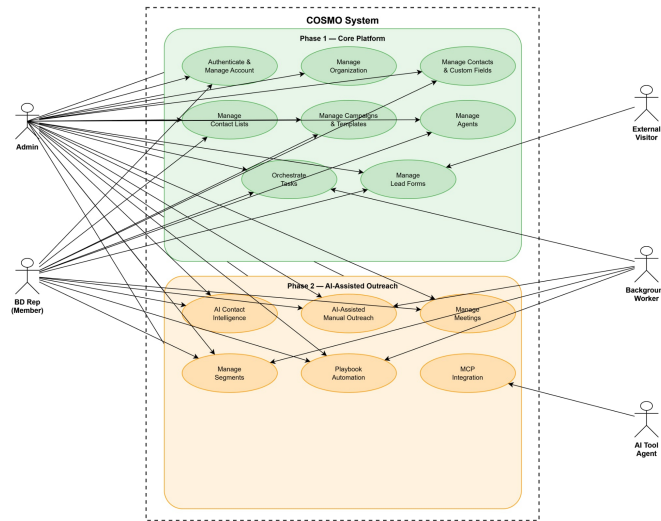


Figure 1: Overall Usecase Diagram

- **AI Tool Agent (MCP):** An external AI tool that calls COSMO APIs via the Model Context Protocol to create a campaign and its associated contacts in a single request.

The automated email outreach workflow is organized into four phases:

1. **Setup:** The BD representative creates contacts, groups them into a contact list, creates a campaign, and assigns an execution agent (with valid Gmail credentials).
2. **Generate:** The system calls the OpenAI API to generate personalized email templates based on the contact context (profile, organization data, interaction history) and the campaign strategy (tone, value proposition, cadence).
3. **Execute:** When the BD representative clicks *Execute*, the backend enqueues tasks into a Redis-backed queue. Async workers poll tasks, load required data from PostgreSQL, and send emails via the Gmail API for each contact. For every send attempt, the worker writes an email log record and updates the task status (e.g., queued → processing → succeeded/failed).
4. **Follow-up:** Workers automatically schedule follow-up tasks according to the configured cadence and update each contact's outreach stage to reflect the latest interaction and next required action.

The manual outreach workflow consists of six steps:

1. **AI Suggest:** The system recommends a prioritized list of contacts to reach out to, based on outreach status (e.g., cold contacts, contacts requiring follow-up) and the time elapsed since the last interaction.
2. **AI Draft:** The BD representative selects a contact, and the system calls the OpenAI API to generate a personalized draft message according to the scenario (e.g., first outreach, no-reply follow-up, post-reply, meeting confirmation) and language (Vietnamese/English).
3. **Manual Send:** The BD representative copies and edits the draft, then sends it manually via LinkedIn, personal email, or phone calls. The system **does not** automatically send messages in this workflow.

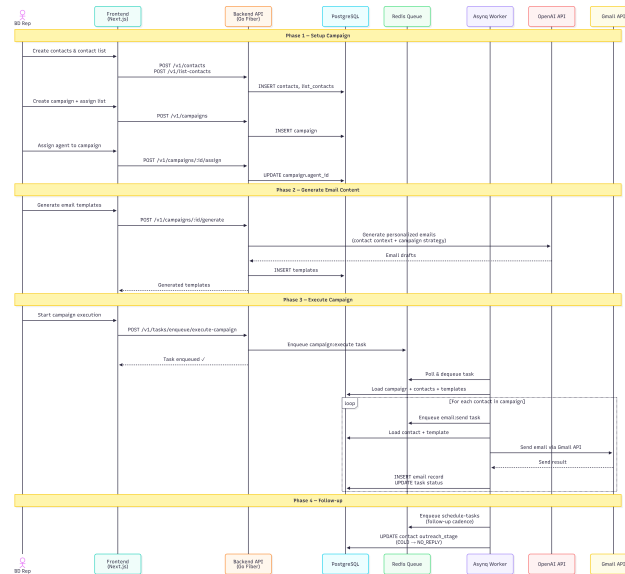


Figure 2: Automated Email Campaign Flow

4. **Log Interaction:** The BD representative records the interaction outcome (channel, direction, sentiment). The system updates the contact outreach state, e.g., COLD → NO_REPLY → REPLIED → POST_MEETING.
5. **Feedback:** The BD representative provides feedback on the AI draft quality (e.g., used as-is, edited, rewritten, skipped), which can be used to improve future suggestions and drafts.
6. **Background Recalculation:** A background worker periodically recalculates outreach status for contacts in a waiting state (e.g., WAIT), automatically updates the next recommended step, and increments follow-up counters where appropriate.

6.2 System Architecture

- **Client Layer:** Users access the system through a web browser.
- **Frontend (Next.js 14):** A React application using the App Router for routing, TanStack React Query for server-state management, and the Ky HTTP client to automatically attach the JWT token to each request. An authentication middleware checks JWTs before allowing access to protected routes.
- **Backend (Go/Fiber v3):** A REST API server following a layered architecture: Middleware → Handler → Service → Repository → Database. The middleware stack handles CORS, JWT authentication, rate limiting, logging, and metrics.
- **Background Workers:** Async workers process asynchronous tasks (e.g., email sending, campaign execution, agent synchronization) from a Redis-backed queue.
- **Data Stores:** PostgreSQL stores primary application data, while Redis is used for caching and as the task queue backend.

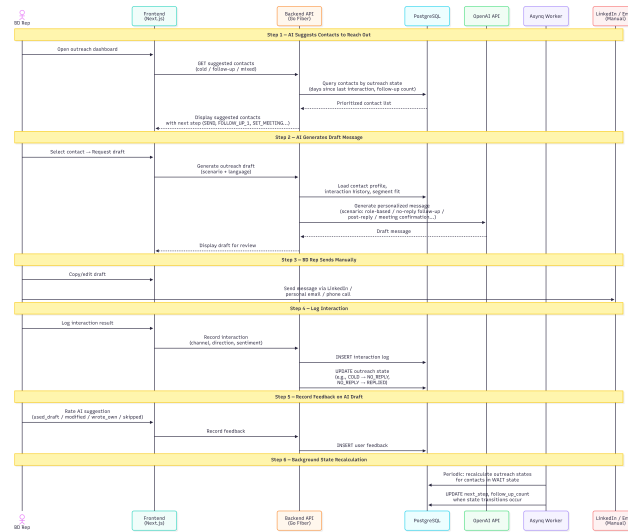


Figure 3: Manual Outreach Flow

- **External Services:** Integrations include Google OAuth (authentication), Gmail API (sending/receiving email), OpenAI (AI content generation), and HubSpot (CRM synchronization).

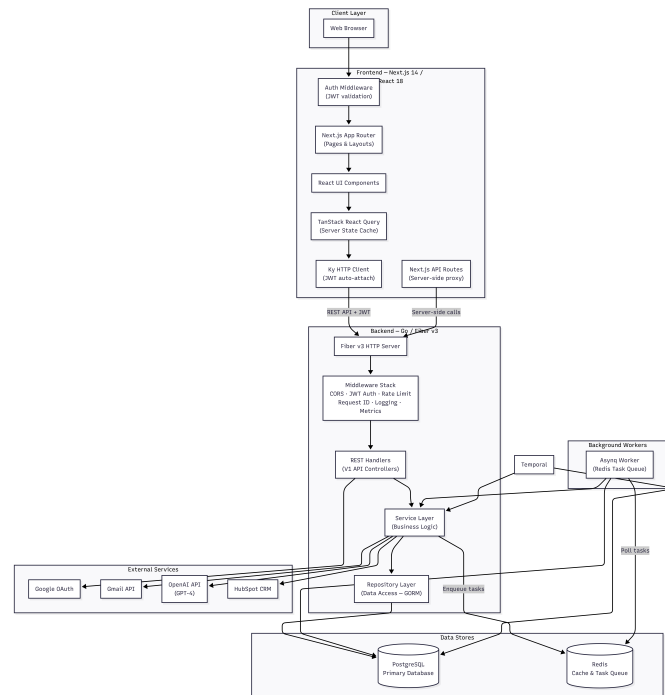


Figure 4: System Architecture Diagram

6.3 System Design

7 Conclusion



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